Fiscal Year:	FY 2007	**Task Last Updated:**	FY 04/04/2007
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Project Title:	Loading-Induce Fluid Flow Mediates Microgravity Related Disuse Bone Loss		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
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Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	94305-3030	**Congressional District:**	14
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No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:	None		
COI Name (Institution):			
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Performance Goal No.:			
Performance Goal Text:			
Task Description:	The bone loss that occurs with disuse in microgravity is a significant challenge that must be overcome in order for long-duration space flights such as a manned mission to Mars to be successful. Recently it has been shown that interstitial fluid flow in bone due to loading is a potent physical signal in regulating bone cell metabolism. The purpose of this project is to determine if this physical signal is also important in the formation of new bone forming cells (osteoblasts). To determine if bone fluid flow due to loading can regulate the formation of new bone cells in terms of proliferation and differentiation of osteoblast precursor cells. We will also determine if the biochemical mechanism involved is similar to the one we have desribed involved in the regulation of mature bone cells.		

	An understanding of the cellular and biochemical mechanism involved in the regulation of bone metabolism due to mechanical loading will result in novel targets for pharmacological intervention, potentially leading to a new class of drugs which could mimic the effects of loading on bone, even in the absence of loading or in the presence of reduced loading.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Disuse bone loss is a critical problem in settings other than microgravity. These include osteoporosis related to age, since inactivity is a known contributing factor, as well as bone loss that occurs with paralysis and stroke.
Task Progress:	Final Report NAG2-1601 The bone loss that occurs with disuse in microgravity is a significant challenge that must be overcome in order for long-duration space flights such as a manned mission to Mars to be successful. Recently it has been shown that interstitial fluid flow in bone due to loading is a potent physical signal in regulating bone cell metabolism. The purpose of this project was to determine if this physical signal is also important in the formation on new bone forming cells (osteoblasts). Thus, the objective of this project was to determine if bone fluid flow due to loading can regulate the formation of new bone cells in terms of proliferation and differentiation of osteoblast precursor cells. The study was conducted with human bone marrow stromal cells (hMSCs) that contain osteoblast precursors and our in vitro cell culture flow system. Outcome variables included proliferation rate determined by BrdU incorporation, PGE2 release, and osteogenic gene expression. Significance: An understanding of the cellular and biochemical mechanism involved in the regulation of bone metabolism due to mechanical loading will result in novel targets for pharmacological intervention potentially leading to a new class of drugs which could mimic the effects of loading on bone even in the absence of loading or in the presence of reduced loading. Our results will also have important terrestrial application. These include osteoporosis related to age since inactivity is a known contributing factor as well as disuse bone loss that occurs with paralysis and stroke. The three year project involved testing the following three hypotheses in three corresponding specific aims: Hypothesis 1: hMSC cells respond to loading-induced oscillatory fluid flow with cytosolic calcium mobilization via a 2nd messenger stimulated release of calcium from intracellular stores. Specific Aim 1: We identified the intracellular calcium signaling pathway activated by loading-induced oscillatory fluid flow. hMSC cells were cultured in vitro and placed in our oscillatory flow system. Cells were exposed to fluid flow and monitored for increased intracellular calcium concentration with a real-time microspectrofluorometer. The flow profile was sinusoidal superposed with a low level of constant wash-through perfusion to provide fresh media. A cell was classified as having responded to fluid flow if its largest during flow calcium oscillation is greater than 4 times the size of its largest pre-flow oscillation. Gadolinium chloride, which blocks stretch activated calcium channels, thapsigargin which empties ER stores of calcium, and nifedipine which blocks voltage activated calcium channels were employed to dissect the calcium response, ie if OFF activates an intracellular or extracellular calcium response and if that response was transduced by stretch activated or voltage activated channels. Thapsigargin significantly decreased the percent of cells which responded to dynamic fluid flow ($p<0.05$) while gadolinium chloride and nifedipine had no effect on the percent of cells responding to fluid flow. While gadolinium chloride and nifedipine had no effect on the percent of cells responding to fluid flow they did significantly decrease the magnitude of the cells calcium response ($p<0.05$). Our results suggest that the mechanisms for the calcium response of MSCs to OFF is the release of intracellular stores of calcium. These data are consistent with what our lab has found in MC3T3-E1 osteoblasts. This consistency suggests a ubiquitous pathway in the transduction of the extracellular signal (OFF) into the cell interior. There also may be a role for the stretch activated and voltage activated calcium response in the magnification of this calcium response, since both gadolinium chloride and nifedipine, while they had no effect on the percent of cells responding, did decrease the magnitude of the calcium response. A manuscript describing these results is currently in preparation and will be submitted in June 2007. Hypothesis 2: Loading-induced oscillatory fluid flow results in increased proliferation and enhanced differentiation of hMSCs down an osteogenic pathway. Specific Aim 2: We determined the effects of exposure to loading-induced fluid flow on the proliferation rate and the expression of markers of differentiation of hMSCs. hMSCs were be exposed to two hours of oscillatory fluid flow with continuous wash-out perfusion and the number of viable cells and proliferation rate determined relative to non-flow exposed controls as a function of time. Expression of markers of differentiation were assayed as a function of time after flow. Markers included quantification of Runx2, type I collagen, osteopontin, and osteocalcin mRNA levels, alkaline phosphatase activity, calcium deposition, and mineralization. We found that mechanical stimulation via fluid flow did indeed increase the proliferation of hMSCs. We further found that osteopontin and osteocalcin mRNA levels were increased with stimulation. However, Runx2 and type I collagen were not. In the case of Runx2, we expect that we did not observe an increase because we did not assay an early enough time-point after flow. In the case of type I collagen, we found very high expression level prior to stimulation and there may not have been sufficient margin for the cells to exhibit an increase. These findings were published in: Li, Y.J., Batra, N.N., You, L., Meier, S.C., Coe, I.A., Yellowley, C.E., Jacobs, C.R., Oscillatory Fluid Flow Affects Human Marrow Stromal Cell Proliferation and Differentiation, Journal Orthopedic Research, 22: 1283-1289, 2004. Hypothesis 3: The effect of loading-induced oscillatory fluid flow on the proliferation and differentiation of hMSCs involves intracellular calcium mobilization. Specific Aim 3: In this experiment we repeated the experiments in SA2 above in the presence of Thapsigargin. Our results are still being analyzed, based on our findings with MC3T3-E1 cells we expect that changes in proliferation and differentiation do indeed involve calcium signaling. These results will be included in the manuscript being prepared from the results of SA1. Other outcomes: In terms of the specifics of this project, we have demonstrated that fluid flow due to loading can increase the proliferation of osteoblast precursors as well as increasing their differentiation into mature osteoblasts. We have also made a related discovery that the dynamics of the time-pattern of loading can have a significant effect on the impact of fluid flow on bone cell metabolism. This may have important implications for optimizing the temporal pattern of mechanical stimulation of bone in physical interventions for the preventions of disuse bone loss. We are seeking

	funding to pursue this idea from the Department of Veterans Affairs. This finding was reported in: Batra, N.N., Li, Y.J., Yellowley, C.E., You, L., Malone, A., Tiberio, C., Jacobs, C.R., Effects of Short Term Recovery Periods on Fluid Induced Calcium Oscillations in Osteoblastic Cells. Journal of Biomechanics, 38:1909-1917, 2005. Funding for this project and the results generated were also described in a review article recently published in Gene about signaling pathways activated my mechanical stimulation in bone: Rubin, J, Rubin, C, Jacobs, CR, Molecular pathways mediating mechanical signaling in bone. Gene, Epub ahead of press, 2006. In addition to regulating the production of bone forming progenitors, mechanical loading may also be involved in regulating the activity of bone resorbing osteoclasts. In this regard the NASA grant allowed us to collect some early data connecting oscillatory fluid flow to regulation of the RANKL/OPG signaling axis. This was published in: Kim CH, You L, Yellowley CE, Jacobs CR. Oscillatory fluid flow-induced shear stress decreases osteoclastogenesis through RANKL and OPG signaling. Bone, Epub ahead of press, 2006. We have begun to build on the results of this project and have begun to search for the molecular mechanism that is responsible for the transduction that activates the signaling pathways described above. Our early work in this area was in determining the role of the cytoskeleton in this process: Malone, A, Batra, NN, Shivaram, G, Kwon, R, You L, Kim CH, Rodriguez, J, Jair, K, Jacobs, CR, The Role of the Actin Cytoskeleton in Oscillatory Fluid Flow Induced Signaling in MC3T3-E1 Osteoblasts. American Journal of Cell Physiology, Epub ahead of print, 2007. Summary: The overall goal of this study was to determine the intracellular mechanisms of the response of hMSCs to OFF in the context of their differentiation into bone and other skeletal tissues. This information will be valuable in the understanding the mechanical environment that promotes hMSCs differentiation and maintenance as osteoblasts. Furthermore, comparing hMSCs and osteoblasts response to OFF will provide insight into the potential specialization of signaling pathways during differentiation. For example, if more signaling pathways are activated in hMSCs than in osteoblasts, some of that generality is lost during differentiation. Or, if the two cell types sense OFF with the same pathway, it implies that this sensitivity is a more fundamental aspect of cell function.

Bibliography Type:	Description: (Last Updated: 02/08/2013)
Articles in Peer-reviewed Journals	Rubin J, Rubin C, Jacobs CR. "Molecular pathways mediating mechanical signaling in bone." Gene. 2006 Feb 15;367:1-16. Epub 2005 Dec 19. Review. PMID: 16361069 , Feb-2006
Articles in Peer-reviewed Journals	Kim CH, You L, Yellowley CE, Jacobs CR. "Oscillatory fluid flow-induced shear stress decreases osteoclastogenesis through RANKL and OPG signaling." Bone. 2006 Nov;39(5):1043-7. Epub 2006 Jul 24. PMID: 16860618 , Nov-2006
Articles in Peer-reviewed Journals	Malone AM, Batra NN, Shivaram G, Kwon RY, You L, Kim CH, Rodriguez J, Jair K, Jacobs CR. "The role of actin cytoskeleton in oscillatory fluid flow-induced signaling in MC3T3-E1 osteoblasts." Am J Physiol Cell Physiol. 2007 May;292(5):C1830-6. Epub 2007 Jan 24. PubMed PMID: 17251324 , May-2007
Articles in Peer-reviewed Journals	You L, Temiyasathit S, Lee P, Kim CH, Tummala P, Yao W, Kingery W, Malone AM, Kwon RY, Jacobs CR. "Osteocytes as mechanosensors in the inhibition of bone resorption due to mechanical loading." Bone. 2008 Jan;42(1):172-9. Epub 2007 Sep 26. PubMed PMID: 17997378 , Jan-2008